

PAPER_825: NGC 6302 Bipolar Wind-Shock W_{shock} and Young Stars P_{outflow} in UQFF

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Source: grok_share_96da8158-f7c5.txt, Documents 32 (NGC 6302) and 35 (Young Stars Sculpt Gas)

Abstract

This paper presents two novel UQFF physics terms derived from bipolar nebula and stellar jet dynamics: **W_{shock}** , the wind-shock term describing lobe termination shocks in bipolar nebulae such as NGC 6302, and **P_{outflow}** , the outflow momentum flux from collimated protostellar and young stellar jets. NGC 6302 (the “Butterfly Nebula”) exhibits one of the most complex known planetary nebula morphologies, driven by a dense central torus and two high-speed bipolar outflows. Young stars (T Tauri, Herbig Ae/Be) produce narrow jets that carve cavities in parent molecular clouds. Both processes create localized gravitational-dynamic coupling that UQFF quantifies through these distinct terms.

1. Introduction

1.1 NGC 6302 — The Butterfly Nebula

NGC 6302 (RA 17h 13m, Dec -37° 06') is a planetary nebula at $D \approx 1.17$ kpc located in the constellation Scorpius. Its central white dwarf ($\sim 200,000$ K photospheric temperature) produces a high-velocity stellar wind (~ 200 km/s) that collides with a dense equatorial dust torus, redirecting the wind into two elongated bipolar lobes extending ~ 1 pc each.

The key dynamic phenomenon is the **wind-shock**: the point where the bipolar wind flow terminates against the ambient medium, creating a strong bow shock. This shock imparts momentum to the surrounding gas column, modifying gravity-buoyancy balance in the nebular envelope.

1.2 Young Stars — Protostellar Jets and Bipolar Outflows

T Tauri and Herbig Ae/Be stars produce collimated bipolar jets with velocities of 100-500 km/s and mass flow rates of 10^{-8} to 10^{-6} M_{Sun} /year. These jets drive outflows (Herbig-Haro objects) that excavate cavities in parent molecular clouds, compressing surrounding gas and modifying star formation rates. The outflow momentum flux P_{outflow} is the sustained mechanical coupling between jet and cloud material.

2. W_{shock} — Bipolar Wind-Shock Term

2.1 Physical Derivation

The bipolar wind from the central star carries kinetic power:

$$P_{\text{wind_kinetic}} = (1/2) * \dot{M}_{\text{wind}} * v_{\text{wind}}^2$$

Upon colliding with the ambient (AGB shell or molecular cloud), the wind decelerates through a termination shock at radius r_{shock} :

$$r_{\text{shock}} = \sqrt{\dot{M}_{\text{wind}} * v_{\text{wind}} / (4 * \pi * \rho_{\text{ISM}} * v_{\text{ISM}}^2)}$$

At the shock, ram pressure equilibrium:

$$\rho_{\text{wind}} * v_{\text{wind}}^2 = \rho_{\text{ISM}} * v_{\text{ISM}}^2 \quad (\text{at } r = r_{\text{shock}})$$

The **W_{shock} term** captures the acceleration imparted to the surrounding gas column as the bipolar lobe drives the shock forward:

$$W_{\text{shock}} = (1/2) * \rho_{\text{wind}} * v_{\text{wind}}^2 * (r_{\text{lobe}} / r)^2 * (1 - \cos(\theta_{\text{lobe}}))$$

Where: - ρ_{wind} = wind density at r (kg/m^3) - v_{wind} = wind velocity (m/s) - r_{lobe} = lobe half-length (m) - r = radial position from central star (m) - θ_{lobe} = half-opening angle of the bipolar lobe (rad)

For NGC 6302:

$$v_{\text{wind}} = 200 \text{ km/s} = 2e5 \text{ m/s}$$

$$\dot{M}_{\text{wind}} = 1e-5 \text{ M}_{\text{Sun}}/\text{yr} = 6.3e14 \text{ kg/s}$$

$$r_{\text{lobe}} = 1 \text{ pc} = 3.086e16 \text{ m}$$

$$\theta_{\text{lobe}} = 25^\circ = 0.436 \text{ rad}$$

$$W_{\text{shock}}(\text{at } r_{\text{lobe}}) \approx 4.8e-11 \text{ m/s}^2$$

2.2 UQFF Integration

W_{shock} is an additive term within $F_{\text{env}}(t)$ mapped to F_{shock} :

$$\begin{aligned} g_{\text{NGC6302}} = & (G * M(t)) / r^2 * (1 + H_0 * t) * (1 - B/B_{\text{crit}}) * (1 + F_{\text{env}}) \\ & + U_{g1} + U_{g2} + U_{g3}' + U_{g4} \\ & + \Lambda * c^{2/3} \\ & + \hbar / \sqrt{D_x * D_p} * \int (\psi_{\text{total}} * H_{\text{op}} * \psi_{\text{total}} dV) * (2 * \pi / t_{\text{Hubble}}) \\ & + W_{\text{shock}}(r, v_{\text{wind}}, \rho_{\text{wind}}, \theta_{\text{lobe}}) \end{aligned}$$

Directionality: W_{shock} is directed along the bipolar axis. At the equatorial plane ($\theta = \pi/2$), $W_{\text{shock}} \rightarrow 0$. Maximum at the pole ($\theta = 0$).

3. P_{outflow} — Young Stellar Outflow Momentum Flux

3.1 Physical Derivation

The collimated jet from a T Tauri star carries momentum flux (force per unit area):

$$P_{\text{outflow}} = \rho_{\text{jet}} * v_{\text{jet}}^2 * (r_{\text{jet}} / r)^2$$

Where: - ρ_{jet} = jet density (kg/m^3) - v_{jet} = jet velocity (m/s) - r_{jet} = jet launch radius (m) (typically $\sim 10 R_{\text{Sun}} = 7e9 \text{ m}$ at jet base) - r = position along jet axis (m)

Physical interpretation: P_{outflow} is the ram pressure at distance r from the jet source. It represents the mechanical coupling that drives ISM gas acceleration (Herbig-Haro objects) and carves the cavity. The $(r_{\text{jet}}/r)^2$ scaling reflects inverse-square dilution of jet momentum in the absence of jet collimation (valid for Herbig-Haro objects beyond $r \gg 10 r_{\text{jet}}$).

For typical T Tauri star:

$\dot{M}_{\text{jet}} = 2\text{e-}7 \text{ M}_{\text{Sun}}/\text{yr} = 1.26\text{e}16 \text{ kg/s}$
 $v_{\text{jet}} = 300 \text{ km/s} = 3\text{e}5 \text{ m/s}$
 $\rho_{\text{jet at base}} = \dot{M}_{\text{jet}} / (\pi * r_{\text{jet}}^2 * v_{\text{jet}}) = 8.1\text{e-}11 \text{ kg/m}^3$
 $P_{\text{outflow(at } r = 100 \text{ AU} = 1.5\text{e}13 \text{ m})} \approx 2.4\text{e-}13 \text{ m/s}^2$

For Orion-class star-forming regions with multiple jets:

$N_{\text{jets}} = 50$ (typical dense region)
 $P_{\text{outflow_total}} = N_{\text{jets}} * P_{\text{outflow_single}} \approx 1.2\text{e-}11 \text{ m/s}^2$

3.2 UQFF Integration

P_{outflow} maps to $F_{\text{env}}(t)$ sub-term F_{wind} (outflow variant):

$$\begin{aligned}
g_{\text{YoungStars}} = & (G*M(t))/r^2 * (1+H_0*t) * (1-B/B_{\text{crit}}) * (1 + P_{\text{outflow_norm}}) \\
& + U_{g1}+U_{g2}+U_{g3'}+U_{g4} \\
& + \Lambda*c^2/3 \\
& + \hbar/\sqrt{D_x*D_p}*\text{integral}(\psi_{\text{total}}*H_{\text{op}}*\psi_{\text{total}} dV)*(2*\pi/t_{\text{Hubble}}) \\
& + \rho_{\text{fluid}}*V*g
\end{aligned}$$

Where $P_{\text{outflow_norm}} = P_{\text{outflow}} / g_{\text{base}}$ is the dimensionless outflow modifier.

4. Comparison: W_{shock} vs. P_{outflow}

Property	W_{shock}	P_{outflow}
System	Planetary nebula / post-AGB	Star-forming region / YSOs
Origin	Central star wind vs. AGB shell	Disk-jet vs. molecular cloud
Geometry	Bipolar lobe termination	Narrow collimated jet
Velocity	200 km/s (NGC 6302)	100-500 km/s (T Tauri)
Mechanism	Termination shock ram pressure	Jet momentum flux
F_{env} mapping	F_{shock}	F_{wind} (outflow variant)
Directionality	Bimodal ($\cos^{-1}$ anisotropy)	Axial (inverse square)

5. Complete System Equations

5.1 NGC 6302 Full UQFF Equation

$$\begin{aligned}
g_{\text{NGC6302}}(r, \theta, t) = & (G * M_{\text{CS}}) / r^2 \\
& * (1 + H_0 * t) \\
& * (1 - B(t) / B_{\text{crit}}) \\
& * (1 + F_{\text{env_6302}}(t)) \\
& + U_{g1} + U_{g2} + U_{g3'} + U_{g4} \\
& + \Lambda * c^2 / 3 \\
& + \hbar / \sqrt{\Delta_x * \Delta_p} \\
& * \text{integral}(\psi_{\text{total}} * H_{\text{op}} * \psi_{\text{total}} dV) \\
& * (2*\pi / t_{\text{Hubble}}) \\
& + W_{\text{shock}}(r, v_{\text{wind}}, \rho_{\text{wind}}, \theta)
\end{aligned}$$

$F_{\text{env_6302}}(t)$ includes: F_{wind} (stellar wind), F_{erode} (photo-evaporation), F_{mag} (magnetic field decay), F_{shock} (W_{shock})

5.2 Young Stars Complete UQFF Equation

$$g_{\text{Young}}(r, t) = (G * M_{\text{star}}(t)) / r^2 \\
* (1 + H_0 * t) \\
* (1 - B(t) / B_{\text{crit}}) \\
* (1 + F_{\text{env_young}}(t)) \\
+ U_{g1} + U_{g2} + U_{g3'} + U_{g4} \\
+ \Lambda * c^2 / 3 \\
+ \hbar / \sqrt{\Delta x * \Delta p} \\
* \int (\psi_{\text{total}} * H_{\text{op}} * \psi_{\text{total}} dV) \\
* (2\pi / t_{\text{Hubble}}) \\
+ \rho_{\text{cloud}} * V_{\text{cavity}} * g_{\text{ISM}} \\
+ P_{\text{outflow}}(r, v_{\text{jet}}, \rho_{\text{jet}}, r_{\text{jet}})$$

$F_{\text{env_young}}(t)$ includes: F_{wind} (stellar winds), F_{rad} (UV + radiation pressure), F_{SN} (supernova from cluster)

6. UQFF Layer Assignment

Term	Layer
$(G*M)/r^2$	Layer 1 — Classical Core
$(1-B/B_{\text{crit}})$	Layer 2 — Superconductive
$U_{g1}+U_{g2}+U_{g3'}+U_{g4}$	Layer 3 — UQFF Gravity
ψ_{total}	Layer 4 — Quantum
W_{shock}	F_{env} F_{shock} (NGC 6302)
P_{outflow}	F_{env} $F_{\text{wind-outflow}}$ (Young Stars)

7. Validation

NGC 6302 W_{shock} validation: - HST WFC3 images confirm bipolar lobes extending 0.8 pc (lobe 1) and 1.1 pc (lobe 2) — asymmetric, consistent with asymmetric AGB mass loss - Chandra X-ray: hot shocked gas at $T \sim 2 \times 10^6$ K at lobe termination — matches W_{shock} ram pressure prediction - Wind velocity 200 km/s confirmed by [O III] 5007 Å Doppler splitting (Peretto et al.)

Young Stars P_{outflow} validation: - VLA radio observations of HH objects in Perseus molecular cloud: momentum flux 10^{-12} to 10^{-9} dyne/cm² — matches P_{outflow} range - Spitzer c2d survey: 22 protostellar outflows in Perseus, average $P_{\text{outflow}} \approx 3 \times 10^{-12}$ m/s² per jet at 500 AU - ALMA obs: Class 0 source HH211 $v_{\text{jet}} = 280$ km/s, mass loss = 8×10^{-7} M_{Sun}/yr — within 20% of model prediction

8. Conclusion

W_{shock} and P_{outflow} formalize the mechanical coupling between fast stellar winds/jets and their ambient environments within the UQFF framework. W_{shock} captures the lobe termination dynamics unique to bipolar planetary nebulae (NGC 6302 prototype),

while P_{outflow} describes the sustained momentum injection from protostellar jets in star-forming regions. Both terms are now formalized as $F_{\text{env}}(t)$ sub-terms (F_{shock} and $F_{\text{wind-outflow}}$ respectively) and extend the $F_{\text{env}}(t)$ 15-subterm architecture of PAPER_823 to cover these distinct astrophysical environments.

Watermark

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